



Review article

Research advances in residual thermal stress of ceramic/metal brazes



Ruixiang Yi^a, Chao Chen^{a b} , Chen Shi^{a b}, Yuxiang Li^b, Haijun Li^a, Yibo Ma^b

Show more

Outline | Share Cite

<https://doi.org/10.1016/j.ceramint.2021.04.220>

[Get rights and content](#)

Full text access

Abstract

Brazing, as a common method of bonding ceramic and metal, has been applied in microelectronics, aerospace, machinery and other domains extensively. The residual thermal stress in the brazed joint has direct effects on the mechanical properties of the joint, so how to control the generation of residual thermal stress has become the vital point. In this paper, the methods of reducing residual thermal stress in the brazing process in recent years are reviewed. The generation and effects of residual thermal stress in the brazed joint are introduced. Besides, the methods of detecting residual thermal stress are discussed, and different methods of reducing residual thermal stress in brazed joints are also analyzed. Finally, the future development directions of reducing residual thermal stress in the brazed joint are proposed.



Keywords

Brazing; Ceramic; Residual thermal stress; Process

1. Introduction

Considering that the concept of science and technology as the first force continues to strengthen, in most cases, modern industry has put forward more stringent requirements for engineering materials. Meanwhile, the single material performance has been unable to meet the needs of our industrial field [\[1\]](#), [\[2\]](#), [\[3\]](#). Owing to its good corrosion properties, high hardness and other excellent performances, ceramic has become an indispensable part of the engineering structures field [\[4\]](#), [\[5\]](#), [\[6\]](#). However, due to its high brittleness and poor processing performance, the single ceramic material is not enough to meet the requirements of the engineering field. In the brazing process, it often needs to be metallurgically connected with metal materials with good plasticity, so that the good performances of ceramics can be played fully [\[7\]](#), [\[8\]](#), [\[9\]](#).

In the connection between ceramic and metal, because of its many advantages, brazing has become a common method [\[10,11\]](#). In a high-temperature vacuum environment, the intermediate filler metal melts, and metallurgy reactions occur between ceramic and metal forming a good metallurgical bonding [\[12,13\]](#). The metallurgy reactions are usually carried out in a high-temperature vacuum environment, which puts forward high requirements to the equipment. However, with the continuous development and progress of technology, the ultrasonic wave is introduced into brazing, and good results are obtained. The cavitation effect of ultrasonic waves can remove the oxide film formed on the substrate surface in the atmospheric environment, and then the full reaction between the filler metal and base metal can be realized. At the same time, the acoustic flow can promote the spread of the molted filler metal on the substrate metal [\[14\]](#), [\[15\]](#), [\[16\]](#), [\[17\]](#). With the aid of ultrasonic waves, the brazing process can be performed not only at lower temperatures but also in atmospheric conditions, which expands its application scope.

However, the inevitable problem is that the thermal expansion coefficient of ceramic and metal in the process of brazing are very different, which will lead to the generation of residual thermal stress in the brazing joint, resulting in the joint cracking and the joint strength reducing [18,19]. The difference in thermal expansion coefficient becomes a great challenge for the brazing between ceramic and metal.

This paper analyzes the causes of residual thermal stress and the influences of residual thermal stress on the bonding strength of joints. Besides, the detection methods of residual thermal stress and the methods of reducing residual thermal stress are introduced. Finally, several development trends of reducing the residual thermal stress of the joint are proposed.

2. The brazing between ceramic and metal

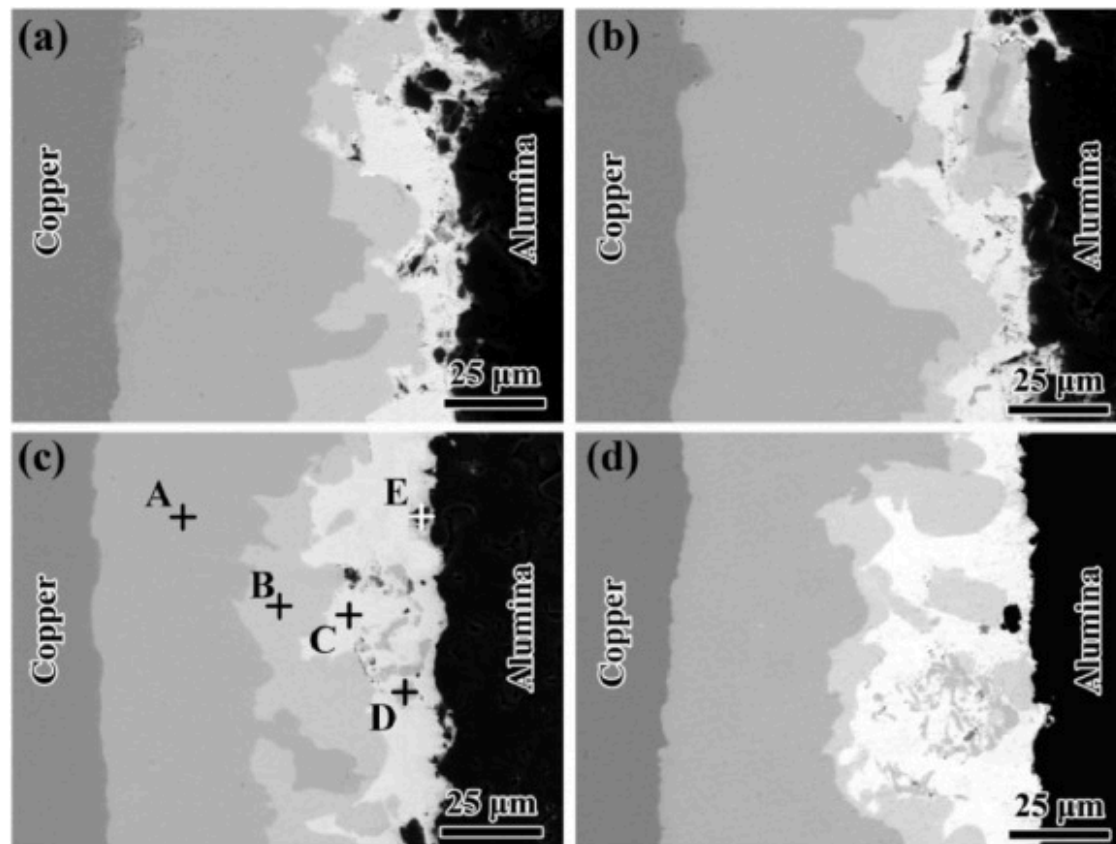
There are many ways to bond the same or dissimilar materials, such as brazing [20, 21, 22], sealing [23], anodic bonding [24] and so on. However, at present, the most commonly used connection is brazing. Due to the different brazing temperature, brazing is divided into soft brazing and hard brazing. In the soft brazing method, there are mainly ultrasonic brazing [25,26], laser brazing [27] and infrared brazing [28,29]. Hard brazing methods mainly include furnace brazing [30], diffusion brazing [31] and induction brazing [32]. Among the numerous brazing methods, furnace brazing, ultrasonic brazing and diffusion brazing are the most commonly used bonding methods between ceramics and metals.

2.1. Active brazing of ceramic and metal

By adding active elements to the filler metal, the reactive layer is formed on the ceramic surface. The chemical reactions in brazing process can enhance the wettability of the filler metal on the ceramic surface to carry out the chemical bonding between ceramic and metal [12,33]. Ti, Zr, and Hf are metal elements in the transition interval of the periodic table. These elements not only have a strong chemical activity but also have a great affinity for oxide and silicate. Meanwhile, these elements are easy to form alloys with Cu, Ni, Ag, and then these liquid alloys can react with ceramic surface. These metal elements are called active elements [34,35].

To explore the influence of active elements in filler metal, Fu et al. [36] studied the effects of Ti content on the bonding strength and microstructure of the joint between Al_2O_3 ceramic and Cu. The content of Ti in filler metal was 2%, 4%, 6% and 8% respectively. The microstructure of brazed joints with different Ti content is shown in Fig. 1. According to Fig. 1 and energy dispersive spectrometer (EDS) measurement results, the phases in the joints were similar, although the Ti content in the joints

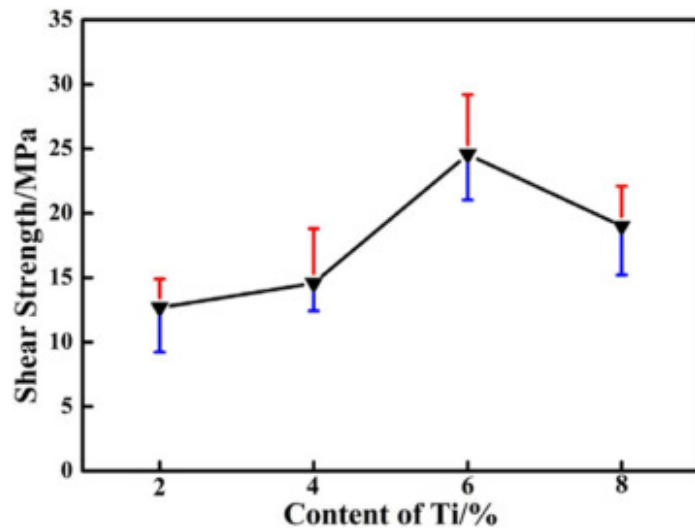
was variable. The reason was that their reaction conditions were the same. In the joints with more Ti content, more grey Ti_6Sn_5 phase would be generated in the weld. The shear test results showed that with the increase of Ti content, the bonding strength of the joint had been greatly improved. This was because the increase of Ti content would increase the wettability of the surface. At the same time, the dispersed Ti_6Sn_5 phase in the joint strengthened the microstructure and prevented the propagation of cracks, thus improving the shear strength of the joint. However, if the Ti content was reached 8%, too much Ti_6Sn_5 phase would weaken the bonding strength, resulting in the low shear strength. The bonding strength curve of joints is shown in Fig. 2. When the content of Ti in the filler metal was 6%, the maximum joint strength existed as 25 MPa. Usually, when the content of active elements is too low, the filler metal has poor wettability to the surface of ceramic, and the bonding strength is low. However, if the content of active elements is too high, it will increase the brittleness of the filler metal and reduce the bonding strength. The high requirements are put forward for continuous exploration and research on the content of active elements in filler metal.



[Download: Download high-res image \(556KB\)](#)

[Download: Download full-size image](#)

Fig. 1. Effect of Ti content on the microstructure of copper/alumina brazed joints (600°C/5 min). (a) 2% Ti, (b) 4% Ti, (c) 6% Ti, (d) 8% Ti [36].



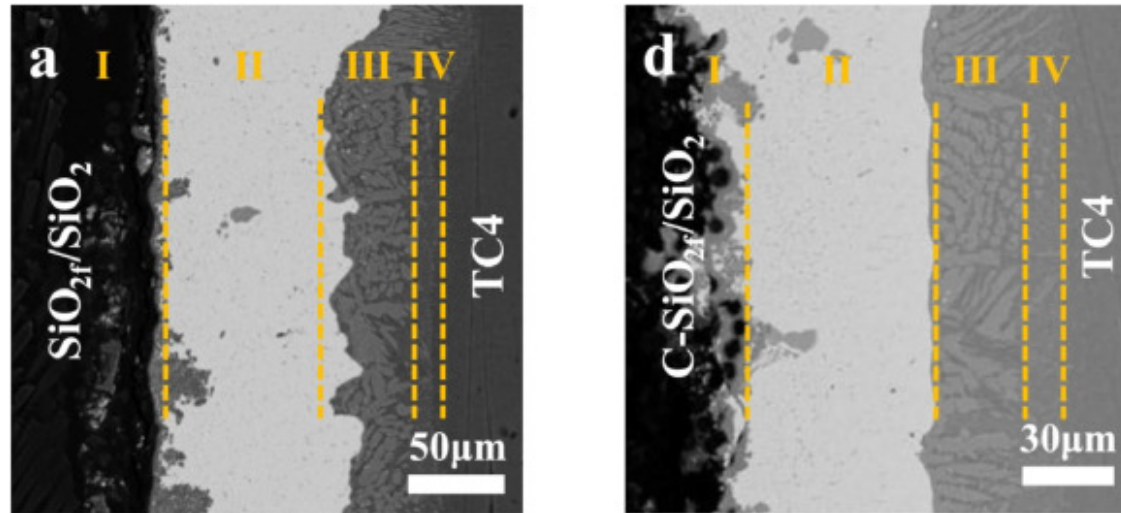
[Download: Download high-res image \(96KB\)](#)

[Download: Download full-size image](#)

Fig. 2. Effect of Ti content on the shear strength of copper/alumina joint brazed at 600°C for 5 min [36].

Lin et al. [37] analyzed the active brazing between $\text{SiO}_{2f}/\text{SiO}_2$ ceramic and TC4 alloy with Ag-26.4Cu-4.5Ti filler metal. The microstructure of the brazed joint is shown in Fig. 3 (a). The joint was mainly composed of four parts as following: the reaction-layers near the base metal, the welding seam and the diffusion layer on the TC4 alloy side respectively. After the mechanical properties test, the average bonding strength of the joint was about 5MPa. To enhance the wettability of the ceramic surface and bonding strength of the joint, plasma treatment was carried out. The $\text{SiO}_{2f}/\text{SiO}_2$ treated by plasma was denoted as C- $\text{SiO}_{2f}/\text{SiO}_2$. The microstructure of the brazed joint after plasma treatment was similar to the joint that without plasma treatment as shown in Fig. 3 (d). The bonding strength after plasma treatment was 7.5MPa, which was higher than the joint strength without plasma treatment on the ceramic surface. The reason was that C- $\text{SiO}_{2f}/\text{SiO}_2$ ceramic had better wettability to the filler metal and the reaction region between C- $\text{SiO}_{2f}/\text{SiO}_2$ ceramic and filler metal in the brazing process was larger. However, the shear strength was

still very low. The reason might be that these Ti–Cu compounds (Ti_2Cu , TiCu) formed in the brazing process were brittle and uneven, and these compounds were not conducive to relieve of residual thermal stress. At the same time, too many Ti elements at the side of TC4 will react violently with $\text{C-SiO}_{2f}/\text{SiO}_2$ and form an over-thick reaction layer. This reaction layer will reduce the joint bonding strength because of its brittleness. Therefore, to significantly enhance the shear strength of the joint, the research direction in the future should be to control the generation and size of metal compounds in the joint.



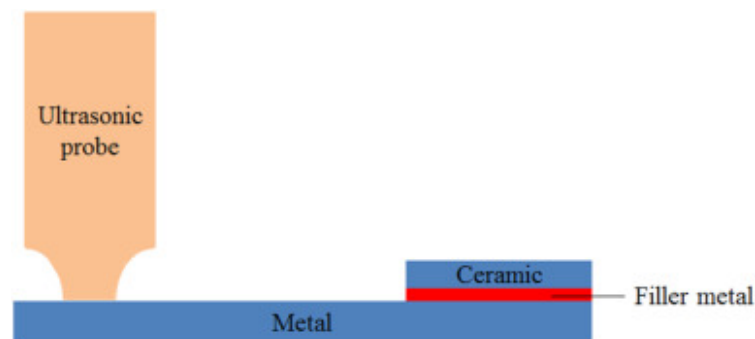
[Download: Download high-res image \(417KB\)](#)

[Download: Download full-size image](#)

Fig. 3. The typical interfacial microstructure of (a) $\text{SiO}_{2f}/\text{SiO}_2$ -TC4 and (d) $\text{C-SiO}_{2f}/\text{SiO}_2$ -TC4 brazed by AgCuTi [37].

2.2. Ultrasonic assisted brazing of ceramic and metal

Ultrasonic assisted brazing is a bonding process in which ultrasonic wave is introduced into brazing process. Ultrasonic vibration not only plays the role of heating, but also plays the role of breaking the surface oxide film. It can replace the flux and realize the flux-free brazing. The effect of ultrasonic wave on the brazing process is mainly to remove the oxide film on metal surface by ultrasonic cavitation effect. When high strength of ultrasonic is introduced into liquid phase, the tiny air bubbles in liquid phase grow under the action of energy. When the energy reaches a certain threshold, these bubbles collapse, and the tremendous energy could destruct the metal surface oxidation film in this process [14,15,38]. Finally, the filler metal is completely spread out on the metal surface to achieve the purpose of welding. The principle of ultrasonic assisted brazing is shown in Fig. 4.



[Download: Download high-res image \(64KB\)](#)

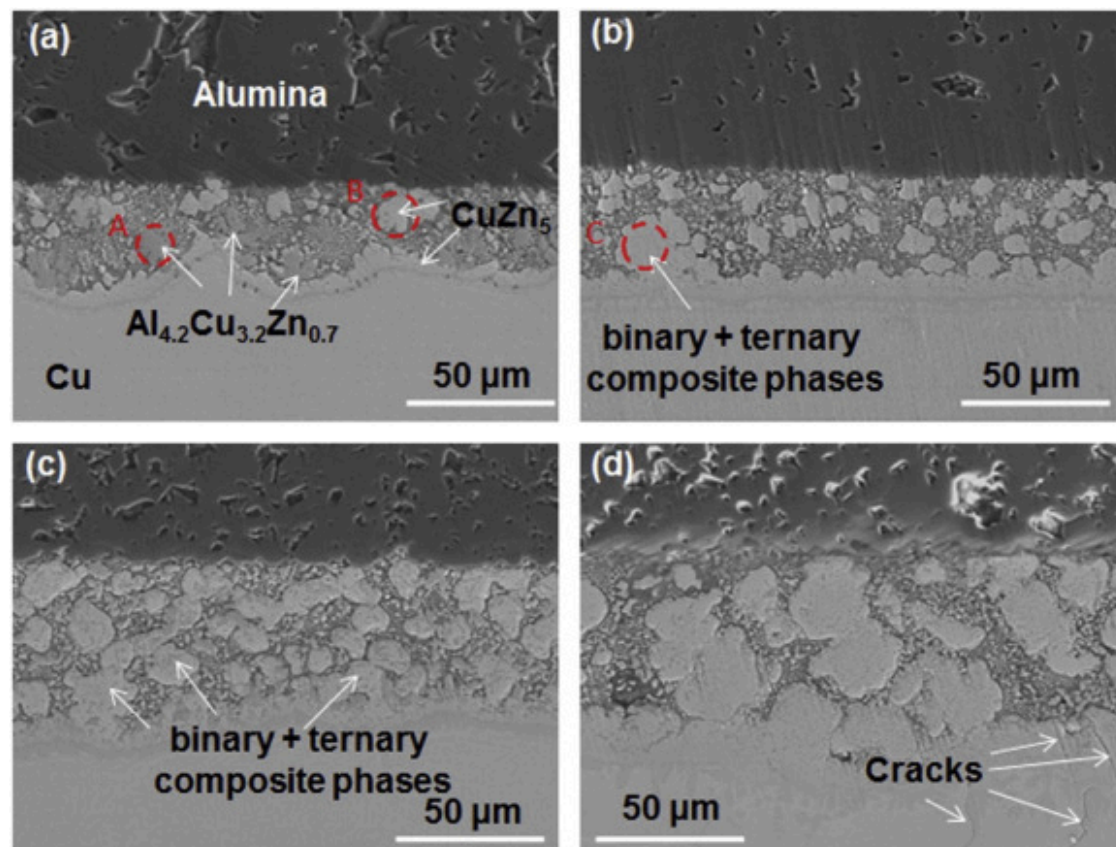
[Download: Download full-size image](#)

Fig. 4. Schematic diagram of ultrasonic brazing principle.

The problem of poor wettability between ceramic and metal can be solved by adding the active elements in filler metal [39]. However, because the active brazing experimental temperature is high, the high residual stress will be formed in the cooling process, which makes the joint deterioration of mechanical properties and reduces the connection strength of joints. So low-temperature brazing becomes the research focus gradually. In the low-temperature brazing methods, one is to remove the oxide film on the surface of the filler metal and the base metal by adding brazing flux, so as to realize the connection. However, the brazing flux is not easy to clean and the brazing flux will remain in the weld, leading to corrosion of parts and reduction of product quality. Another method is to apply ultrasonic wave in the brazing process. In the process of brazing, the energy of the ultrasonic wave is applied to remove the oxide film formed on the surface of the substrate, so that the base metal has good wettability without considering the problem of active elements [14,15]. At the same time, ultrasonic-assisted brazing can realize the effective connection between ceramic and metal in a medium or low temperature environment. When the brazing temperature was 360°C, Roman et al [40] found that the connection between AlN ceramic and Cu was realized using Zn–Al–Mg filler metal with the assistance of ultrasound, and the maximum shear strength reached 47MPa. Meanwhile, Nake et al. [41] succeeded in brazing between Al₂O₃ ceramic and Cu with the assistance of ultrasound, and the brazing temperature was 450°C. Therefore, the ultrasonic assisted brazing has a better development prospect.

Ji et al. [42] investigated the brazing of Al₂O₃ ceramic and Cu using Zn–Al filler metal with the assistance of ultrasonic waves. The melting point temperature of Zn–14Al filler metal was 380°C. They discussed the influences of different ultrasonic times on the

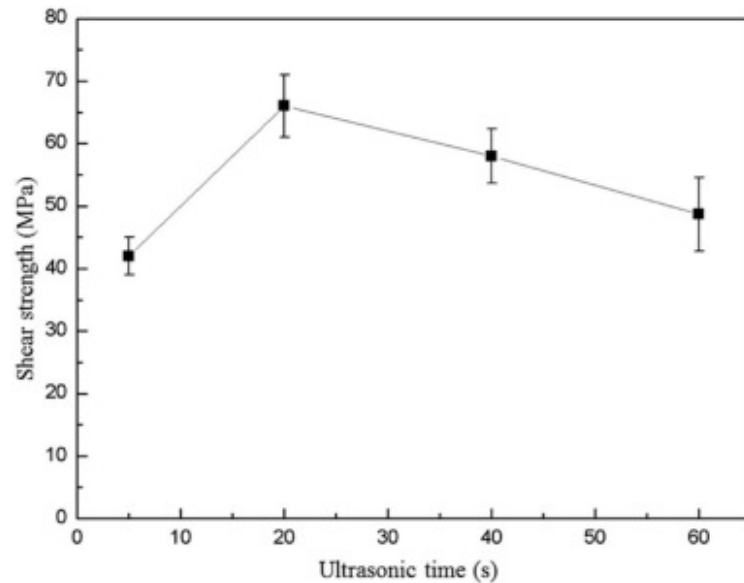
joint's microstructure and bonding strength. The joint microstructure under different ultrasonic times is shown in Fig. 5. It can be seen that from Fig. 5, the joint interface was complete without obvious defects. When the ultrasonic time was 5s, there was a dendritic gray metal compound layer CuZn_5 on the Cu side. With the increase of ultrasonic time, the metal compound layer became thicker, and the ternary Al–Cu–Zn phase could continue to grow. The reason was that the extension of ultrasonic time would increase the solubility of Cu. By the mechanical properties testing, the bonding strength curve of the joint was a parabolic shape. The reason was that if the ultrasonic time was too short, the filler metal and the base metal were not sufficiently reacted, resulting in low shear strength. When the ultrasonic time was too long for the 60s, the brittle metal compound layer became thicker, which also resulted in the decline of shear strength. When the ultrasonic vibration time was the 20s, as shown in Fig. 6, the maximum bonding strength can be obtained about 66MPa. Ultrasonic-assisted brazing is better than active brazing to some extent. However, because of its many variables, such as ultrasonic time and ultrasonic amplitude, the microstructure and bonding strength of the joint were susceptible to these factors. So it is very important to further study the influencing mechanisms of these variables on the joint for obtaining the optimal joint strength.



[Download: Download high-res image \(792KB\)](#)

[Download: Download full-size image](#)

Fig. 5. SEM images of the joints that were ultrasonically brazed for different ultrasonic vibration duration times of (a) 5s, (b) 20s, (c) 40s, and (d) 60s [42].



[Download: Download high-res image \(93KB\)](#)

[Download: Download full-size image](#)

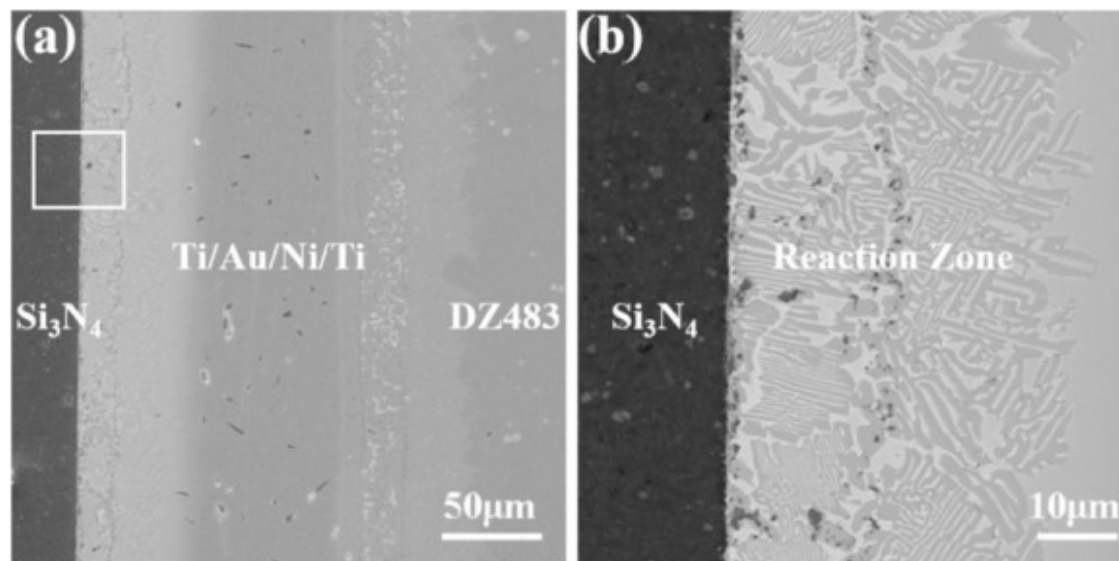
Fig. 6. Shear strength of the brazed $\text{Al}_2\text{O}_3/\text{Zn-14Al}/\text{Cu}$ joints as a function of ultrasonic vibration duration time [42].

2.3. Transient liquid phase (TLP) brazing of ceramic and metal

TLP brazing is a connection method by presetting filler metal or coating in the joint gap and then heating filler metal until it is liquid phase, so that the filler metal can be partially or completely eliminated and finally the complete joint can be obtained [43,44]. In a gas or vacuum environment, the intermediate layer is melted into a liquid phase by increasing the temperature, while a certain pressure is applied to the sample surface. The liquid metal will fill the gap on the base metal surface, and element diffusion will occur between the solid and liquid phases to forming a dense intermediate transition layer, thus achieving the purpose of welding [45,46].

Due to the uniform composite and structure of the weld formed by TLP brazing, the joint has high strength and good plasticity, which is widely used in the connection of same or dissimilar materials [47,48].

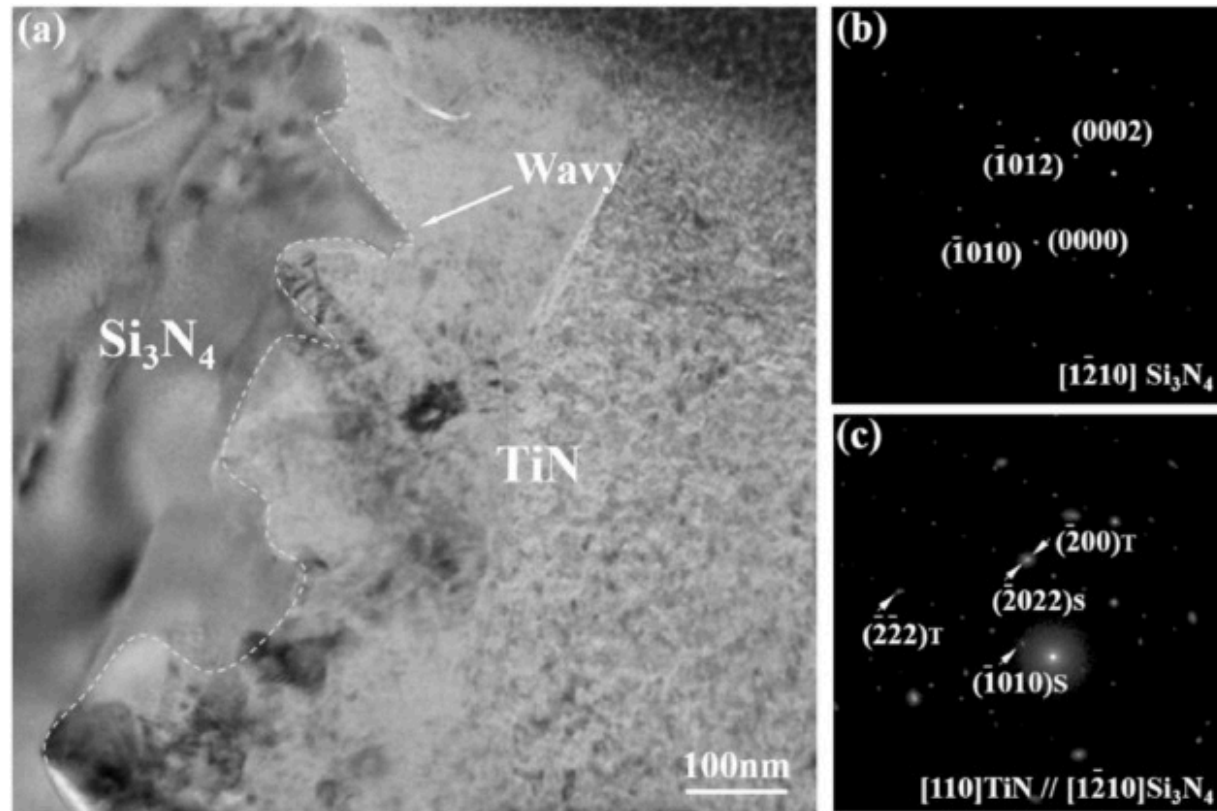
Lan et al. [47] investigated the bonding between Si_3N_4 ceramic and nickel-base superalloy (DZ483) by TLP bonding using Ti/Au/Ni interlayers. In their previous studies, Ti/Cu/Ni composite interlayers could effectively connect the ceramic to the metal, but failure usually occurred in the filler metal because Cu was easy to oxidize in the high temperature environment. However, Au interlayer is a good alternative to Cu interlayer because of its strong oxidation resistance and good chemical stability. Meanwhile, its plastic deformation can relieve the residual thermal stress caused by the difference of thermal expansion coefficient. When Ti/Au/Ni composite interlayer was used, the connection was complete without obvious defect, as shown in Fig. 7. It can be seen from Fig. 7 that there were obvious reaction regions on the ceramic side, which were composed of Au-rich phase, Ni-rich phase with gray layered structure and reaction layer TiN in the Si_3N_4 ceramic side. By enlarging the ceramic side area further, it was found that interface between the Si_3N_4 ceramic and the reaction layer TiN presented a wavy shape, as shown in Fig. 8. The reason was that the formation of TiN particles and the growth direction of TiN particles were random, which led to the wavy ceramic interface. Unfortunately, Lan et al. did not test the mechanical properties of the bonding samples. Whether the wavy surface of the ceramic will improve the bonding strength of the joint and the fracture mode of the bonding samples are unknown, and these problems still need to further research and discovery. Meanwhile, because of the high cost of Au, this approach may not be suitable for practical applications.



[Download: Download high-res image \(375KB\)](#)

[Download: Download full-size image](#)

Fig. 7. SEM images of the Si_3N_4 /nickel-base superalloy joint bonded at 1273K for 60min: (a) general view; (b) Si_3N_4 /Ni interface [47].



[Download: Download high-res image \(548KB\)](#)

[Download: Download full-size image](#)

Fig. 8. (a) TEM morphology at Si_3N_4 /TiN interface; (b) SADP of the Si_3N_4 ; (c) shows $(\bar{1}\bar{1}1)\text{TiN}/(\bar{1}010)\text{Si}_3\text{N}_4$, $(\bar{1}00)\text{TiN}/(\bar{1}011)\text{Si}_3\text{N}_4$ (the suffixes “T” and “S” denote TiN and Si_3N_4 , respectively) [47].

For the brazing between ceramic and metal, these methods have to face an inevitable question. The question is the generation of the residual thermal stress in the cooling process. Reducing the brazing temperature can appropriately ease the residual thermal stress of the brazing joint to some extent. Considering that the difference of thermal expansion coefficient between metal and ceramic is large, the residual thermal stress will inevitably affect the bonding strength of the joint and even seriously make the

joint failure. Therefore, in addition to reducing the experiment temperature, there should be other ways to transit the thermal expansion coefficient between ceramic and metal as well as decrease the generation of residual thermal stress in joint to get high-quality brazing joint.

3. The residual thermal stress in the joint

Considering that the coefficient of thermal expansion between ceramic and metal is different, the deformation of ceramic and metal is different in the cooling process, which will lead to the residual thermal stress in the brazed joint. If the residual thermal stress is greater than the joint strength, it will lead to the failure and destruction of the joint interface [49]. In the case of strong bonding strength, the residual stress often causes the fracture of ceramics directly. Therefore, understanding the distribution and magnitude of residual thermal stress near the joint surface is greatly significant for the development of high-strength ceramic-metal joints.

3.1. Generation of residual thermal stress

Parts will deform under the action of external forces, resulting in strains and stresses within the parts. However, the deformation of an object is not only caused by external effects but also caused by temperature change. The deformation by temperature change is called thermal deformation [50]. It needs to be pointed out that the temperature change alone does not necessarily produce stress in the object. When the expansion or contraction caused by the temperature change is constrained, stress will be produced in the object. Due to temperature changes, this kind of constraints, which is not subjected to external force but is caused by the thermal deformation, is called the thermal stress or temperature stress [[51], [52], [53]]. The stress is caused by non-external force, and the fundamental cause of thermal stress is temperature change and constraint action. The restraint action can be summarized into three forms, as following: the constraint of external deformation, the constraint of mutual deformation and the constraint of deformation among internal parts [54].

In the brazing process, residual thermal stress is generated in the joint owing to the shrinkage difference between ceramics and metals during cooling [55]. The brazing residual thermal stress can be divided into longitudinal residual thermal stress and transverse residual thermal stress according to the direction relative to the weld [56]. The longitudinal residual thermal stress is expressed as parallel to the weld direction, and the transverse residual thermal stress is expressed as perpendicular to the weld direction. The longitudinal residual stress is mainly tensile stress, and its peak value can reach the yield strength of the material

[57]. Because the ceramic deformation is small in cooling, the ceramic is mainly subjected to pressure stress. To make the joint internal stress balance, the ceramic is mainly affected by the longitudinal residual stress.

3.2. Test methods for residual thermal stress

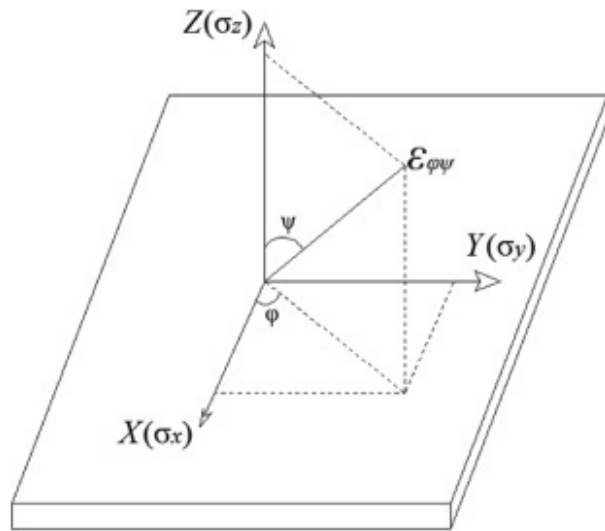
Residual thermal stress testing methods have always been the focus of attention. With the improvement of the research level and testing instruments, the residual thermal stress detection methods have become very diverse. The measuring principle of residual thermal stress varies with the method of residual stress detection. At present, the commonly used detection methods include the X-ray and neutron diffraction method [58], hole-drilling method [59], contour method [60], nano-indentation method [61] and so on. The types and characteristics of residual thermal stress detection methods are shown in Table 1.

Table 1. Common methods of residual stress detection.

	Types	advantages	disadvantages
X-ray diffraction method	Non-destructive	Widespread use	Easily affected by material structure
Neutron diffraction method	Non-destructive	High accuracy	High cost expensive equipment
Nano-indentation method	Non-destructive	Micron scale	Small measuring area
Hole-drilling method	Semi-destructive	Normalization, Widespread use	Damaging the samples
Ring-core method	Semi-destructive	Measuring coarse-grained materials	Not standardized, Long measurement process
Sectioning method	Destructive	Simple operation High accuracy	Damaging the samples
Contour method	Destructive	Simple test low cost	Damaging the samples

X-ray method is a non-destructive technique of measuring material surface residual stress. When metal is subjected to external or internal stresses, the elastic strain can lead to the change of metal atoms spacing in the crystal structure. Meanwhile, X-ray diffraction (XRD) can directly measure the interatomic distance between planes and then calculate the residual stress value of the material surface [62,63]. However, the range of residual stress measured by the XRD method is within 10 μ m of the material surface, but the accuracy of the XRD method will be greatly reduced when there is a certain texture in the material or the

microstructure of the welded joint is not uniform. The schematic diagram of the X-ray diffraction method is shown in Fig. 9. The principle of the neutron diffraction method is roughly the same as that of the X-ray diffraction method, but the neutron diffraction method has not been widely used because of its high cost and expensive equipment [64].

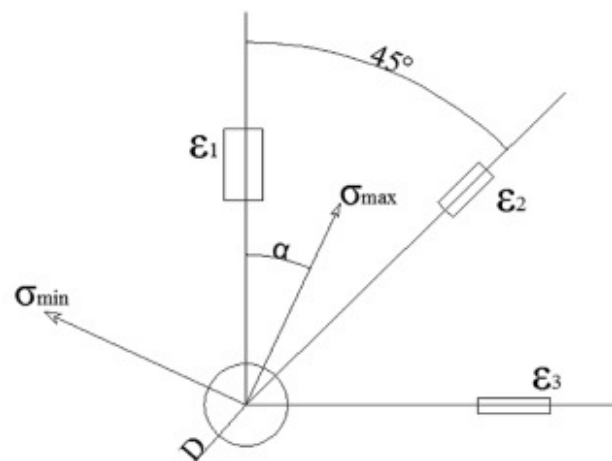


[Download: Download high-res image \(88KB\)](#)

[Download: Download full-size image](#)

Fig. 9. Schematic diagram of residual stress measurement by XRD.

Compared with other measurement methods, the Hole-drilling method is more practical and simpler. As one of the most widely used semi-destructive methods, the Hole-drilling method can measure the distribution of residual stress in the whole material. Meanwhile, the characteristics of the Hole-drilling method are high reliability and good accuracy [65,66]. The principle is that the residual stress of the keyhole is calculated by the strain on the surface of the keyhole, as shown in Fig. 10. In the process of measuring residual stress, however, there are elastic modulus error, patch error and other error factors. At present, these error factors are analyzed and researched in the existing literature, but these analyses are not comprehensive, which makes it impossible to comprehensively improve the residual stress measurement accuracy and can not fully guarantee the reliability of the measurement results.



$\epsilon_1, \epsilon_2, \epsilon_3$ - Strain values at different angles
 $\sigma_{\max}, \sigma_{\min}$ - Residual stress value

[Download: Download high-res image \(123KB\)](#)

[Download: Download full-size image](#)

Fig. 10. Schematic diagram of residual stress measurement by Hole-drilling method.

In the process of nano-indentation testing, when there is compressive stress on the material surface, the actual contact area of indentation is the largest [67,68]. By the load-displacement curve of the indentation point, the residual stress in the material could be calculated. Nano-indentation can measure the variation of residual stress in the material on the micron scale, and other measurement methods only can measure the minimum in the millimeter scale [69]. However, due to the small area of indentation, the difference between different phases in the multiphase material can easily affect the measurement results.

3.3. Influence of residual thermal stress on mechanical properties of joint

During the preparation of ceramic-metal brazing joints, the residual stress was generated as the experimental temperature cooled to room temperature. These residual thermal stresses can significantly reduce the bonding strength of the brazed joint and even lead to the damage and failure of joint interface in the process of brazing. Chen et al. [70] investigated the ultrasonic-assisted brazing between SiC ceramic and TC4 alloy with different filler metals. When the filler metal Al-12Si was used, the joint cross-section was in good contact with no defects, but the crack appeared in the SiC ceramics and expanded in a direction

parallel to the brazing weld. However, when the Al-15.5Sn-9.5Si-4.5Zn-0.5Mg filler metal was used, no cracks were found on the ceramic side. Fig. 11 shows the macroscopic morphology of different solder welds. The thermal expansion coefficient of SiC ceramic is $4.1 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$ and that of the Ti-6Al-4V alloy is $9.1 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$. Considering that the coefficient of thermal expansion between ceramic and metal is different, residual thermal stress will be produced in joints in the cooling process. When the temperature dropped, the joints using Al-Si filler metal had a strong resistance to deformation. When the interface formed a stronger bond, the brittle ceramics were more sensitive to the coefficient of thermal expansion and were in a fragile state. With the continuous decrease of experimental temperature, the residual thermal stress increased and finally led to the crack on the ceramic side. However, when Al-Sn-Si-Zn-Mg filler metal was used, the shrinkage freedom of SiC ceramics and Ti-6Al-4V alloy was large before solidification of liquid Sn-rich phase, which was not easy to cause cracks in ceramics. The shear test results showed that the bonding strength of the joint with cracks in ceramic was low, with a maximum bonding strength of 35.2MPa. However, the maximum bonding strength of the joint with no cracks was 83.5MPa, which indicated that the cracks in ceramic would seriously affect the joint strength to cause the safety accidents in the practical applications.

[Download: Download high-res image \(532KB\)](#)

[Download: Download full-size image](#)

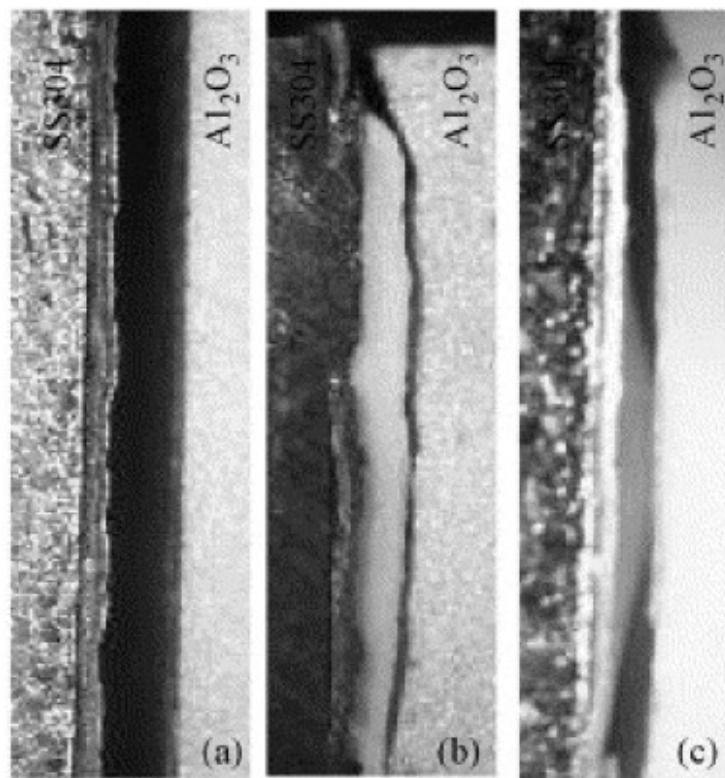
Fig. 11. Macroscopic morphology (LM) of the cross-sections of SiC/Ti-6Al-4V joints filled with (a) AlSi and (b) AlSnSiZnMg alloys [70].

Zhang et al. [71] analyzed the influences of residual thermal stress on brazing joints of the same or dissimilar materials. They found that under the same conditions, the bonding strength between Al_2O_3 ceramic was higher than that brazing between Al_2O_3 ceramics and SS304 stainless steel. The reason for this phenomenon was that the thermal expansion coefficient of ceramic and metal was very different, as shown in Table 2. From Table 2, the thermal expansion coefficient of metal is about 2.5 times that of ceramic, and the thermal expansion coefficient increases non-linearly with the increase of temperature. Due to the difference of thermal expansion coefficient between metal and ceramic, the metal and ceramic degree of shrinkage will be different in the cooling process, which can produce residual thermal stress in the joint. The residual thermal stress can cause metal plastic deformation, ceramic internal crack or interface debonding. Those above defects would lead to the decrease of the strength of the joint even premature failure. Bending tests were performed on the connection samples, and there were three different fracture forms, as shown in Fig. 12. However, due to the high residual thermal stress in the joint, the crack occurred first on the ceramic

side and finally expanded along the ceramic side. The huge difference in thermal expansion coefficient will produce high residual thermal stress, which will greatly deteriorate the mechanical properties of the joint and bring huge economic losses to the actual production and application.

Table 2. Properties of materials at different temperatures [71].

Temperature (°C)	Alumina			Stainless steel-304		
	CTE (x10 ⁻⁶ , °C)	Elastic Modulus (GPa)	Poisson's ratio	CTE (x10 ⁻⁶ , °C)	Elastic Modulus (GPa)	Poisson's ratio
<400	8.002	372	0.27	20.187	210	0.29
400–600	8.578	372	0.27	21.147	179	0.29
600–800	9.154	372	0.27	22.107	170	0.29
800–1000	9.73	372	0.27	23.067	160	0.29
1000–1200	10.306	372	0.27	24.027	150	0.29
1200–1300	10.594	372	0.27	24.507	142	0.29



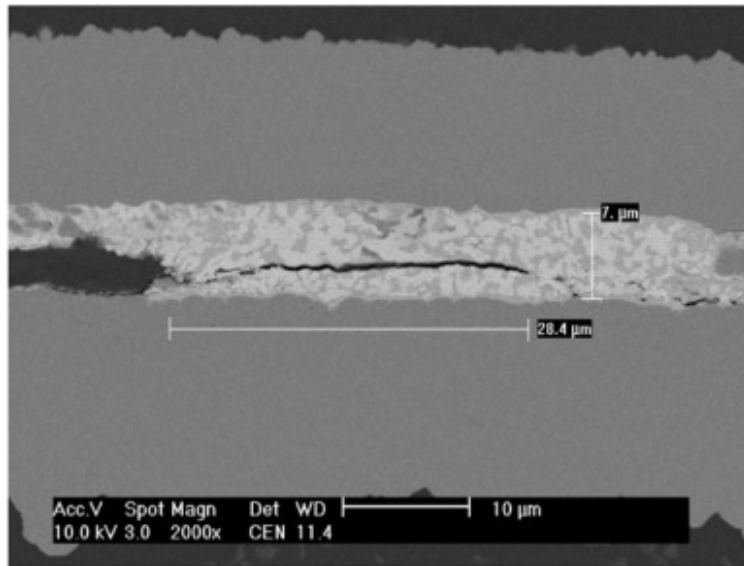
[Download: Download high-res image \(566KB\)](#)

[Download: Download full-size image](#)

Fig. 12. Three different types of fracture in the bending tests [71].

In order to improve the mechanical properties of the joint, Zhang et al. [72] studied the mechanism of solder joint cracks. In the welding of flex-on-board (FOB), the filler metal was the anisotropic conductive films (ACFs), and the ACFs were composed of polymer resin and conductive particles. They found that during the welding process, when the pressure on the surface of the sample was removed, the reaction force generated by the polymer resin would affect the welded joint without the protection of pressure due to its viscoelastic properties. At this time, the filler metal was in a molten state and can not limit the rebound of the polymer, thus leading to the generation of crack in the solder joint. The solder joint crack in the welded joint is shown in Fig. 13. Therefore, the elastic properties of polymer resins had a significant effect on the solder joints. In order to obtain complete joints

and stable mechanical properties of the solder joints, they proposed to use ultrasonic assisted and add silica fillers in the polymer resin, and achieved good results.



[Download: Download high-res image \(283KB\)](#)

[Download: Download full-size image](#)

Fig. 13. Solder joint cracks after a thermo-compression bonding [72].

Residual thermal stress has a great influence on the use of connection sample, especially in brittle fracture, deformation, corrosion resistance, dimensional stability and fatigue life [73]. Considering that the coefficient of thermal expansion is different between metal and ceramic, the value of residual thermal stress in the joint is high, which leads to the failure of the brazed joint. Reducing the residual thermal stress of the joint has become an urgent problem to be solved. At present, many researchers, from home and abroad, have used X-ray diffraction and finite element analysis methods to study the state and distribution of residual stress in ceramic-metal brazing joints and have achieved certain results.

4. Methods for reducing thermal residual thermal stress in brazing

The bonding of ceramic and metal can form complementary advantages, which is widely used in the fields of mechatronics, aerospace, biomedicine and so on [74,75]. Because of its unique advantages and economic feasibility for materials with different bonding properties and melting sensitivity, brazing is a widely used technology in the industrial field [76,77]. During the cooling process of the brazed joint, the coefficient of thermal expansion between ceramic and metal is different, which causes the incongruous thermal deformation and the production of residual thermal stress. With the continuous progress of technology, methods to reduce residual stress tend to be diversified, as shown in Fig. 14. To reduce the residual thermal stress in the brazed joint, many researchers have carried out a lot of research and achieved certain results, mainly from the following three aspects to relieve the residual thermal stress of the brazed joint, as shown in Table 3.

[Download: Download high-res image \(458KB\)](#)

[Download: Download full-size image](#)

Fig. 14. The development trend of methods to reduce residual stress.

Table 3. Brazing of ceramic to metal with different pretreatment methods.

Types	Welded structure	Process parameter	Reaction product or interface structure	Shear or tensile strength/MPa	References
Laser treatment	SiO ₂ /Ag35Cu2Ti/304 stainless steel	Vacuum 2×10 ⁻³ Pa 900°C	The interfacial reaction region is grooved	66MPa	[80]
Adding interlayer	SiO ₂ -BN/AgCuTi/Cu/AgCu/Invar alloy	Vacuum 2×10 ⁻³ Pa 1123 k	Cu(s,s), Ag(s,s) TiN+TiB ₂	43MPa	[7]
	Al ₂ O ₃ /AgCuTi/foam/AgCuTi/stainless steel	Air 800°C		33MPa	[87]
Using composite filler metal	Si ₃ N ₄ /AgCu _c /TC4 alloy	Vacuum 2×10 ⁻³ Pa 880°C	Si ₃ N ₄ /TiN+Ti ₅ Si ₃ /Ag(s,s)+TiCu ₂ /Ti–Cu reaction layers/TC4	73.9MPa	[91]

4.1. Pretreatment of brazing

The residual thermal stress between metal and ceramic joints is closely related to the structure of the joints. If the ceramic surface can be redesigned locally [78], such as laser drilling [79], the periodic uniform distribution of grooves can be formed on the ceramic surface, which will help to alleviate the stress of the joint. Zhang et al. [80] used femtosecond laser to fabricate micron-scale periodic grooves on ceramic surfaces, and the brazing between ceramic and stainless steel with Ag-Cu-Ti filler metal was studied. They used lasers of different power levels to process the ceramic surface, as shown in Fig.15. The experimental results showed that the joint strength of the untreated ceramic surface was 24 MPa, while the joint strength of the grooved ceramic surface can reach 66 MPa, increased by about 2.75 times. For the fracture forms of the joint, the former crack formed at the edge of the weld firstly and then extended to the edge of the ceramic. In the latter case, the crack first extended along with the joint interface and then into the interior of the ceramic. The result of finite simulation showed that the periodic surface grooves formed by femtosecond laser could change the residual stress. The sparse grooves made the interface residual stress change suddenly from the tensile state to the compression state, which restrained the crack growth and enhanced the bonding strength of the joint.

[Download: Download high-res image \(323KB\)](#)

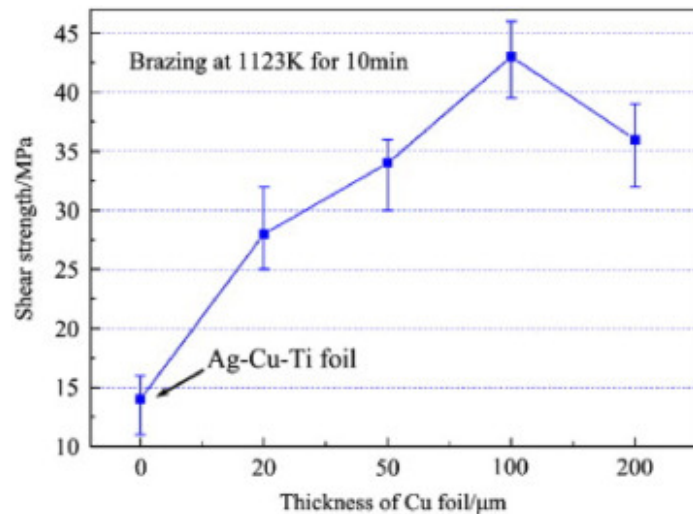
[Download: Download full-size image](#)

Fig. 15. Schematic diagram of the 3D structure of (a) original ceramic surface and the surfaces after (b) 0.3W and (c) 3.4W laser ablation [80].

The addition of an interlayer is also effective in controlling the residual thermal stress of the dissimilar material joint, and it is often used to reduce the residual thermal stress of ceramic-metal brazed joints [81]. The main types of interlayer are soft interlayer, hard interlayer, composite interlayer, porous material and metal fiber mesh buffer layer [[82], [83], [84]]. Usually, the soft interlayer (Cu, Al, Ni, etc.) has a high thermal expansion coefficient, small elastic modulus, low yield point and good plasticity. It can relieve residual stress by its yield, plastic deformation and creep. However, the rigid interlayer (W, Mo) has a large elastic modulus. The small thermal expansion coefficient of the rigid interlayer not only can reduce the difference of thermal expansion coefficient in the joint but also transfer the residual thermal stress concentration area from the weld side of the ceramic to the interlayer, which effectively prevents the initial crack from starting in the ceramic. In a certain size range, the larger the thickness of the intermediate layer is, the smaller the thermal residual stress is. However, if the thickness of the

intermediate layer exceeds a certain value, the influence of increasing the thickness on reducing the residual thermal stress of the joint will be very small. Compared with single interlayer, double layer or multilayer buffer interlayer has better effects on reducing residual thermal stress of brazed joint [85,86], which is especially suitable for ceramic and metal joints with a large difference in thermal expansion coefficient.

To reduce the residual thermal stress of the brazed joint of SiO₂-BN ceramic and Invar alloy, the Cu foil as the interlayer was used. Yang et al. [7] found that if the Cu foil was added to the brazing joint, the properties of the brazed joint with Cu foil was greatly improved. The effects of Cu foil thickness on the brazed joints were also discussed. They found that as the thickness of Cu foil increasing, the wavy compounds Fe₂Ti and Ni₃Ti in the brazed joint gradually decreased. When the thickness of Cu foil was 100μm, these compounds completely disappeared, which was beneficial to improve the mechanical properties of the joint. Meanwhile, with the continuous increase of Cu foil thickness, the bonding strength increased first and then decreased. When the thickness was 100μm, the maximum bonding strength was obtained about 42MPa, which was higher than the bonding strength without Cu foil, as shown in Fig. 16. Restraining the production of brittle compounds and promoting the formation of fine solid solutions were the main reasons for the improvement of joint strength.



[Download: Download high-res image \(218KB\)](#)

[Download: Download full-size image](#)

Fig. 16. Effect of Cu-foil thickness on the shear strength of the joint [7].

Shirzadi et al. [87] investigated the brazing of Al_2O_3 ceramic and stainless steel using Ag–Cu–Ti filler metal, and the stainless steel foam as the interlayer material. After brazing, the photos of the joint interlayer are shown in Fig. 17. From Fig. 17, it can be seen that the eutectic Ag–Cu filler metal had immersed in the stainless steel foam, so the interlayer became a composite structure. The bonding strength of the joint was up to 33MPa, and the brazed joint can be able to stand more than 60 thermal shocks between 200°C and 600°C in the air. The method of using interlayer to reduce residual thermal stress can improve joint performance and enhance joint mechanical strength effectively. Meanwhile, this method has been widely used in brazing between metal and ceramic and has become the main method to relieve residual thermal stress of joint.

[Download: Download high-res image \(542KB\)](#)

[Download: Download full-size image](#)

Fig. 17. Intermediate layer microstructure of alumina and stainless steel brazed joint [87].

As one of the common methods to reduce residual thermal stress, the composite filler metal is also widely used [88,89]. The common measures to reduce the thermal expansion coefficient of the solder layer are directly adding particles to the conventional filler metal or generating the reinforcing phase with a low thermal expansion coefficient in the soldering process [90]. By adding nano- Si_3N_4 and Ti particles into Ag–Cu filler metal (AgCu_C), Zhao et al. [91] reduced the thermal expansion coefficient of the weld and enhanced the joint strength successfully. Fig. 18 shows the joint microstructure of different filler metals. From Fig. 18, the thickness of the reaction layer near the metal side was significantly reduced, which meant that the addition of nanoparticles could effectively restrict the growth of metal compounds TiN and Ti_5Si_3 . At the same time, when AgCu_C filler metal was used, the Ag-based solid solution in the weld was replaced by the Ag-based reinforcing phase. These reinforcing phases distributed on the Ag base can not only improve the strength of the weld but also reduce the difference of the thermal expansion coefficient on the weld. Compared with the filler metal without nano- Si_3N_4 particles, the maximum bonding strength of the brazed joint with composite filler metal was 74MPa, which was about 20MPa higher than that without composite filler metal. The difference of thermal expansion coefficient of base metal can be reduced by adding nano-ceramic particles to the filler metal, and the change of thermal expansion coefficient in a step-like way also can reduce the residual thermal stress of the joint and enhance the mechanical properties of the joint effectively.

[Download: Download high-res image \(367KB\)](#)

[Download: Download full-size image](#)

Fig. 18. The typical joint microstructure of (a) AgCu_C filler metal and (b) AgCu filler metal [91].

4.2. Treatment in the brazing process

To alleviate residual thermal stress of brazing joint, adding interlayer in filler metal, using composite filler metal and treating base metal are effective. Meanwhile, welding temperature, heating rate and cooling rate also can be optimized to improve welding quality and bonding strength of joints [92].

Mishra et al. [93] studied the effects of sample geometry and thermal cycling on residual stress in brazed joints by finite element simulation analysis. Al₂O₃ ceramic and Kovar alloys were ring type, and their dimensions are shown in Table 4. By the finite element analysis, it was found that the residual thermal stress in the brazed joint was reduced with the increase of the base metal size. The reason was that the residual stress reduced with the increase of the curvature diameter. Besides, the brazed joints were treated with different thermal cycles. The first type of brazing joints cooled naturally after brazing, and the second brazing joints were circulated between 200°C and 500°C for 5 times and then cooled naturally to room temperature. The last brazing joints were circulated between 200°C and 500°C for 10 times and finally cooled to room temperature. The results showed that the heating cyclic and cooling cycle can reduce the residual thermal stress caused by the different thermal expansion coefficients in the cooling process. At the same time, the larger sample size is, the lower residual thermal stress of the joint is, and the more obvious the alleviation effect is. Under the condition of a certain sample size, thermal cycling can relieve the residual stress in the brazed joint to some extent and improve the quality of the weld, which provides a positive guiding role for us in the test.

Table 4. Dimensions of the various geometries [93].

Dimensions of various geometry	Inner Diameter (mm)	Outer Diameter (mm)	Kovar Thickness (mm)	Filler Thickness (mm)	Alumina 1 thickness (mm)	Alumina 2 thickness (mm)
Geometry No.1	50	60	1	0.03	5	10
Geometry No.2	70	90	1	0.03	5	10
Geometry No.3	100	120	1	0.03	5	10

Kar Abhijit et al. [94] analyzed the brazing of alumina to stainless steel with Ag–Cu–Ti filler meal and studied the mechanical properties and microstructure of the brazed joints at different temperatures. Experimental results showed that the Al_2O_3 ceramic side was mainly made up of TiO , Al_2TiO_5 , and $\text{Cu}_3\text{Ti}_3\text{O}$ phases, while the stainless steel side consisted of FeTi and $\text{Fe}_{35}\text{Cr}_{13}\text{Ni}_3\text{Ti}_7$. Since the thermal expansion coefficient of Al_2TiO_5 phase was low, the strain of Al_2TiO_5 phase generated in the brazing process can effectively reduce the generation of residual thermal stress in the joint and improve the joint strength. If the brazing temperature reached 900°C , the ceramic side of the joint was composed of TiO and $\text{Cu}_3\text{Ti}_3\text{O}$, and no Al_2TiO_5 phase was produced. These phases led to a large residual tensile stress on the ceramic side and reduced bonding strength. The shear strength was 64MPa by the shear tests. However, when brazing temperature was 1000°C , the rigid phase of Al_2TiO_5 was the most obvious, which effectively reduced the strain energy and increased the joint strength to 94MPa. While the brazing temperature further reached 1100°C , an indistinguishable metal reaction layer appeared at the brazing interface as shown in Fig. 19 (c). Because of the brittleness of the metal reaction layer formed by the metal compound, the residual thermal stress in the joint was large, so the joint strength was greatly reduced. The maximum bonding strength was 46MPa. With the change of brazing temperature, the reaction products in the joint would also be different. The characteristics of the reaction products are different, which will change the residual thermal stress in the joint, thus affecting the bonding strength of the brazing joint. Therefore, in the following research, the main research direction is the shape, nature and arrangement of the phase in the reactants, and these parameters may directly or indirectly affect the joint strength.

[Download: Download high-res image \(638KB\)](#)

[Download: Download full-size image](#)

Fig. 19. SEM–BS images of the Al_2O_3 –304SS brazed interface. Different reaction product layers can be clearly distinguished from the contrasts. From left to right, sample brazed at (a) 900°C ; (b) 1000°C ; (c) 1100°C [94].

In the process of brazing, by treating the base metal and filler metal or adjusting the test parameters, the residual thermal stress in the joint can be effectively reduced to obtain the brazing joint with better performance and high bonding strength. However, to adapt to the development of the times and the progress of science and technology, the application and development of new functional materials have become the main research focus. It still has great research value to study the methods of reducing the residual thermal stress for the brazing connection between different new functional materials.

5. Areas for future work

The brazing process is one of the common methods for bonding metal and ceramic. However, residual thermal stress in the brazing joints is an inevitable problem. To reduce the generation of residual stress in the joint and improve the joint quality, many researchers have done a lot of studies and have achieved certain results, which promote the development of dissimilar materials brazing. However, at present or in the future, it is still a difficult problem to alleviate the residual stress for brazing of dissimilar materials, and there is still a large space for development. Future direction may include the following:

- (1) The application range of ultrasonic-assisted low-temperature brazing should be expanded further.

High-temperature active brazing can obtain good joint strength. However, due to its high residual stress, active brazing can not meet the requirements of practical application, so some researchers put forward the idea of low-temperature ultrasonic-assisted brazing [95,96]. Low-temperature brazing refers to using filler metal with a low melting point and the low melting point filler metal mainly is Sn-based filler metal (Sn–Zn filler metal). However, at present, low-temperature ultrasonic-assisted brazing is mainly used in brazing of ceramic–ceramic [97] or between metal and metal [98]. There are few reports about the brazing of dissimilar materials, such as ceramic to metal. Meanwhile, Sn-based filler metal also has many shortcomings, such as poor oxidation resistance, poor brittleness and low bonding strength [99,100]. To deal with these problems and enhance the wettability, oxidation resistance and mechanical properties of the filler metal, some researchers try to add some alloy elements in Sn base filler metal, such as Al, Cu, Ag, etc [101,102]. However, the reaction mechanism of elements and base metal is still a worth studying problem to reduce the production of metal compounds in the joint and expand the connection of dissimilar materials by adding alloy elements in filler metal.

- (2) Post brazing heat treatment of ceramic brazing joints is carried out.

Both the pretreatment of the material before brazing and the control of test parameters during the brazing process can relieve the residual thermal stress in the joint to a certain extent [28,103]. However, there are few reports on the post-treatment of ceramic and metal brazing joints. The heat treatment process after brazing is mainly for the joint between metal and metal. Post brazing treatment mainly includes annealing treatment, solid solution treatment and aging treatment. After certain annealing treatment, the residual thermal stress of the joint decreases, and the grain size of the weld grows up. Meanwhile, the morphology tends to be uniform and the compound phase in the weld decreases [104]. After proper solution treatment and

aging treatment, the compound phases of grain boundary disappear. The grain boundary becomes flatter, and the micro-hardness and the bonding strength of the joint are improved obviously [105]. Niu et al. [106] studied the effects of brazing temperature for 7075 alloy brazing joints. By mechanical properties testing, they found that with the increase of temperature, the bonding strength of the joint showed a downward trend. When the temperature was too high, the grain size in the joint became coarser, and the shear strength was reduced. To solve the problem of grain enlargement, the brazed joint was treated with a solid solution. After the solution treatment at 490°C and solution time for 1.5h, significant results were obtained in grain refinement, and the uniform fine microstructure was obtained in the joint. To improve the bonding strength of the joint, retrogression and reaging treatment (RRT) was carried out. After heat treatment, the metal compounds in the brazed joint will be dissolved in the matrix and form η' phase or η phase, which could prevent the movement of dislocations, thus improving the bonding strength of the joint. Fig. 20 shows the bonding strength curve of the joint before and after heat treatment. From Fig. 20, it can be seen that the bonding strength of the brazed joint after heat treatment has been greatly improved, which means that the post-heat treatment process after brazing has a significant impact on reducing the residual thermal stress and improving the bonding strength of the joint. However, the current post brazing heat process is mainly applied to the same materials, and the connection between dissimilar ceramic and metal is rarely reported, which may be a hot research direction in the future.

(3) Two or more composite processes can be used to reduce residual thermal stress.

[Download: Download high-res image \(276KB\)](#)

[Download: Download full-size image](#)

Fig. 20. (a) The effect of retrogression temperature on shear strength and (b) the shear strength of joints before and after post-weld heat treatment (PWHT) [106].

The residual thermal stress of the brazed joint can be alleviated to a certain extent by treating base metal and filler metal or adjusting test parameters as described in the previous section. In the cases where the composite filler metal and structural design on the surface of the substrate are used at the same time, it is not clear that the effects of reducing residual thermal stress in the brazed joint is better or not, which still needs to be followed by continuous research and exploration.

6. Summary

The residual stress generated in the brazing process will worsen the mechanical properties of the joint and cause huge economic losses in practical application. Therefore, it has become an urgent problem to relieve the residual thermal stress of joints. The researchers have proposed the drilling-hole method, the nano-indentation method and other methods to detect the distribution and magnitude of residual thermal stress in the joint. To relieve the residual thermal stress of joints, researchers have also put forward many methods from materials to process and have achieved some results.

At present, active brazing is the main way of connecting between ceramic and metal. Although this approach can obtain a steady connection joint, the experiment is carried out at high temperatures, and the experiment shall be performed in a vacuum or inert gas condition, which needs the equipment with more sophisticated and complex. Because the brazing temperature is higher than ultrasonic-assisted brazing, the larger thermal stress will be produced in the joint in the cooling process, which is a difficult point in the process of brazing. By the combination of simulation and experiment, we can clearly understand the distribution and numerical value of residual stress in the joint. Therefore, by changing the test parameters or treating the filler metal and base metal, the residual thermal stress in the brazing joint is greatly reduced, in line with the requirements of practical application.

In the future, ultrasonic-assisted brazing has better application prospects compared with active brazing. Besides, the research and development of new filler metals should be accelerated. High melting point filler metals, such as Ag base filler metal, are not suitable for the brazing of low melting point alloys to ceramics. Therefore, the research and development of low melting point filler metal have a great significance to expand the application of low melting point alloys. Finally, the accuracy of the residual stress detection method could be improved. There are complex compounds in the brazing joint of ceramic and metal, and the physical properties of these compounds are not easy to be measured accurately, which makes the residual thermal stress in the joint uncertain obtained by simulation. Therefore, the study of reliable residual stress measurement methods is also a major research direction.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research work is supported by the National Natural Science Foundation of China (Grant No. [51805416](#)), Young Elite Scientists Sponsorship Program by CAST, Natural Science Foundation of Hunan Province (Grant No. [2020JJ5716](#)), the Project of State Key Laboratory of High Performance Complex Manufacturing, Central South University (Grant No. [ZZYJKT2019-01](#)), Huxiang High-Level Talent Gathering Project of HuNan Province (Grant No. [2019RS1002](#)), and Fundamental Research Funds for the Central Universities of Central South University (Grant No. [1053320192410](#)).

[Recommended articles](#)

References

- [1] Y.Y. Song, H.L. Li, H.Y. Zhao, D. Liu, X.G. Song, J.C. Feng
Interfacial microstructure and mechanical property of brazed copper/SiO₂ ceramic joint
Vacuum, 141 (2017), pp. 116-123
 [View PDF](#) [View article](#) [Google Scholar](#) ↗
- [2] H. Bian, Q. Zhang, Y. Song, D. Liu, J. Liu, X. Song
Microstructure and mechanical properties of a SiO₂ ceramic and TC4 alloy joint brazed with a nanocomposite filler
J. Mater. Eng. Perform., 28 (2019), pp. 4427-4433
[Crossref](#) ↗ [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [3] R.X. Yi, C. Chen, Y.X. Li, H. Peng, H.Y. Zhang, X.Q. Ren
The bonding between glass and metal
Int. J. Adv. Manuf. Technol., 111 (2020), pp. 963-983
[Crossref](#) ↗ [View in Scopus](#) ↗ [Google Scholar](#) ↗
- [4] Z.W. Yang, J.H. Yang, Y.J. Han, Y. Wang, D.P. Wang
Microstructure and mechanical properties of 17-4PH stainless steel and Al₂O₃ ceramic joints brazed with graphene reinforced Ag-Cu-Ti brazing alloy

Vacuum (2020), p. 181



[View PDF](#)

[View article](#)

[Google Scholar ↗](#)

- [5] L. Sun, Q. Wei, J. Fang, C. Liu, J. Zhang
Brazing of porous Si₃N₄ ceramic to Invar alloy with a novel Cu–Ti filler alloy: microstructure and mechanical properties

Ceram. Int., 47 (2021), pp. 2068-2076



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [6] L.X. Zhang, Z. Sun, Q. Chang, J.M. Shi, J.C. Feng
Brazing SiO₂f/SiO₂ composite to Invar alloy using a novel TiO₂ particle-modified composite braze filler

Ceram. Int., 45 (2019), pp. 1698-1709



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [7] Z.W. Yang, L.X. Zhang, Y.C. Chen, J.L. Qi, P. He, J.C. Feng
Interlayer design to control interfacial microstructure and improve mechanical properties of active brazed Invar/SiO₂–BN joint

Mater. Sci. Eng., A, 575 (2013), pp. 199-205



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [8] Z. Sun, L.X. Zhang, T.D. Hao, J.C. Feng
Brazing of SiO₂f/SiO₂ composite to Invar using a graphene-modified Cu-23Ti braze filler

Ceram. Int., 44 (2018), pp. 15809-15816



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [9] C. Shi, R.X. Yi, C. Chen, H. Peng, X.K. Ran, S.D. Zhao
Forming mechanism of the repairing process on clinched joint

J. Manuf. Process., 50 (2020), pp. 329-335



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [10] Y.H. Zhou, D. Liu, H.W. Niu, X.G. Song, X.D. Yang, J.C. Feng

Vacuum brazing of C/C composite to TC4 alloy using nano-Al₂O₃ strengthened AgCuTi composite filler

Mater. Des., 93 (2016), pp. 347-356



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [11] T. Lin, M. Yang, P. He, C. Huang, F. Pan, Y. Huang
Effect of in situ synthesized TiB whisker on microstructure and mechanical properties of carbon–carbon composite and TiBw/Ti–6Al–4V composite joint
Mater. Des., 32 (2011), pp. 4553-4558
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [12] S. Hu, D. Feng, L. Xia, K. Wang, R. Liu, Z. Xia, H. Niu, X. Song
Joints of continuous carbon fiber reinforced lithium aluminosilicate glass ceramics matrix composites to Ti60 alloy brazed using Ti-Zr-Ni-Cu active alloy
Chin. J. Aeronaut., 32 (2019), pp. 715-722
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [13] A.B.D. Sciti, L. Esposito
Bonding of zirconia to super alloy with the active brazing technique
J. Eur. Ceram. Soc., 21 (2001), p. 8
[View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] Z. Li, Z. Xu, L. Ma, S. Wang, X. Liu, J. Yan
Cavitation at filler metal/substrate interface during ultrasonic-assisted soldering. Part I: cavitation characteristics
Ultrason. Sonochem., 49 (2018), pp. 249-259
 [View PDF](#) [View article](#) [Google Scholar ↗](#)
- [15] Z. Li, Z. Xu, L. Ma, S. Wang, X. Liu, J. Yan
Cavitation at filler metal/substrate interface during ultrasonic-assisted soldering. Part II: cavitation erosion effect
Ultrason. Sonochem., 50 (2019), pp. 278-288

[View PDF](#)[View article](#)[View in Scopus ↗](#)[Google Scholar ↗](#)

- [16] E.S.D.G. Shchukin, V. Belova, H. Möhwald
Ultrasonic cavitation at solidsurfaces

Adv. Mater., 23 (2011), pp. 1922-1934

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [17] Y. Xiao, M. Li, L. Wang, S. Huang, X. Du, Z. Liu
Interfacial reaction behavior and mechanical properties of ultrasonically brazed Cu/Zn–Al/Cu joints

Mater. Des., 73 (2015), pp. 42-49



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [18] Z.W. Yang, L.X. Zhang, W. Ren, M. Lei, J.C. Feng
Interfacial microstructure and strengthening mechanism of BN-doped metal brazed Ti/SiO₂–BN joints

J. Eur. Ceram. Soc., 33 (2013), pp. 759-768



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [19] P.L.A. Zanchett, P. Lefort
Ceramic to metal sealings: interfacial reactions mechanism in a porcelain-kovar junction

J. Alloys Compd., 228 (1995), pp. 86-95

[Google Scholar ↗](#)

- [20] S. Zhang, Q. Wang, T. Lin, P. Zhang, P. He, K.-W. Paik
Cu–Cu joining using citrate coated ultra-small nano-silver pastes

J. Manuf. Process., 62 (2021), pp. 546-554



[View PDF](#)

[View article](#)

[Google Scholar ↗](#)

- [21] Q. Wang, S. Zhang, T. Lin, P. Zhang, P. He, K.-W. Paik
Highly mechanical and high-temperature properties of Cu–Cu joints using citrate-coated nanosized Ag paste in air

Prog. Nat. Sci.: Mater. Int., 31 (2021), pp. 129-140

[View PDF](#)[View article](#)[Crossref ↗](#)[Google Scholar ↗](#)

- [22] J. Xu, D. Gao, X. Wang, X. Zhou, B. Zhu, X. Qi, D.P. Sekulic, P. He, Z. Li, T. Lin, S. Zhang
Joining ZnS ceramics by using PbTiO₃-doped PbO–B₂O₃–ZnO
J. Mater. Sci., 55 (2020), pp. 8814-8828
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [23] T.S. Chern, H.L. Tsai
Wetting and sealing of interface between 7056 Glass and Kovar alloy
Mater. Chem. Phys., 104 (2007), pp. 472-478
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [24] D. Xiong, J. Cheng, H. Li, W. Deng, K. Ye
Anodic bonding of glass–ceramics to stainless steel coated with intermediate SiO₂ layer
Microelectron. Eng., 87 (2010), pp. 1741-1746
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] Y. Wu, H. Li, W. Qu, J. Zhang, H. Zhuang
Ultrasonic-assisted bonding of Al₂O₃ ceramic, Cu, and 5056 aluminum alloy with Sn–Zn–Sb solders
Weld. World, 64 (2019), pp. 247-256
[Google Scholar ↗](#)
- [26] R.X. Yi, C. Chen, Y.X. Li, C. Shi, X.Q. Ren
Brazing of SiO₂ ceramic
Int. J. Adv. Manuf. Technol., 113 (2021), pp. 1799-1816
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [27] T.H.I. Südmeyer, M. Rohde
On the shear strength of laser brazed SiC–steel joints: effects of braze metal fillers and surface patterning
Ceram. Int., 36 (2010), p. 8
[Google Scholar ↗](#)

- [28] P. Dong, S. Song, J. Zhang
Analysis of residual stress relief mechanisms in post-weld heat treatment
Int. J. Pres. Ves. Pip., 122 (2014), pp. 6-14
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [29] Y.X. Li, C. Chen, R.X. Yi, Y.W. Ouyang
Review: special brazing and soldering
J. Manuf. Process., 60 (2020), pp. 608-635
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [30] J. Lin, J. Ba, Y. Cai, Q. Ma, D. Luo, Z. Wang, J. Qi, J. Cao, J. Feng
Brazing SiO₂/SiO₂ with TC4 alloy with the help of coating graphene
Vacuum, 145 (2017), pp. 241-244
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] N.R.J. Hynes, P.S. Velu, R. Kumar, M.K. Raja
Investigate the influence of bonding temperature in transient liquid phase bonding of SiC and copper
Ceram. Int., 43 (2017), pp. 7762-7767
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] M. Viljus, V. Vainola, J. Kübarsepp, A. Laansoo
Induction brazing of cermets to steel
Est. J. Eng., 18 (2012)
[Google Scholar ↗](#)
- [33] H. Bian, Y. Song, D. Liu, Y. Lei, X. Song, J. Cao
Joining of SiO₂ ceramic and TC4 alloy by nanoparticles modified brazing filler metal
Chin. J. Aeronaut., 33 (2020), pp. 383-390
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] H.Y. Chen, X.C. Wang, L. Fu, M.Y. Feng

Effects of surface microstructure on the active element content and wetting behavior of brazing filler metal during brazing Ti₃SiC₂ ceramic and Cu

Vacuum, 156 (2018), pp. 256-263



[View PDF](#)

[View article](#)

[Google Scholar ↗](#)

[35] H. Xiong, X. Li, W. Mao, Y. Cheng

Wetting behavior of Co based active brazing alloys on SiC and the interfacial reactions

Mater. Lett., 57 (2003), pp. 3417-3421



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[36] W. Fu, S.P. Hu, X.G. Song, C. Jin, J.X. Li, Y.X. Zhao, J. Cao, G.D. Wang

Effect of Ti content on the metallization layer and copper/alumina brazed joint

Ceram. Int., 43 (2017), pp. 13206-13213



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[37] J.H. Lin, D.L. Luo, S.L. Chen, D.S. Mao, Z.Y. Wang, Q. Ma, J.L. Qi, J.C. Feng

Control interfacial microstructure and improve mechanical properties of TC4-SiO₂f/SiO₂ joint by AgCuTi with Cu foam as interlayer

Ceram. Int., 42 (2016), pp. 16619-16625



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[38] Y. Li, C. Chen, R. Yi

Recent development of ultrasonic brazing

Int. J. Adv. Manuf. Technol., 114 (2021), pp. 27-62

[Google Scholar ↗](#)

[39] J.H. Yang, L.X. Zhang, Z. Sun, J.C. Feng

Wetting and reaction of BN-SiO₂ ceramic by molten Ti-40Ni-20Nb (at.%) filler alloy

Appl. Surf. Sci., 499 (2020)

[Google Scholar ↗](#)

- [40] R. Kolenak, I. Kostolny, J. Drapala, P. Zackova, M. Kuruc
Direct ultrasonic soldering of AlN ceramics with copper substrate using Zn–Al–Mg solder
Metals, 10 (2020)
[Google Scholar ↗](#)
- [41] K.M. Hafez, M. Naka
Applying of ultrasonic waves on brazing of alumina to copper using Zn-Al filler alloy
J. Mater. Sci., 38 (2003), p. 4
[Google Scholar ↗](#)
- [42] H. Ji, H. Chen, M. Li
Overwhelming reaction enhanced by ultrasonics during brazing of alumina to copper in air by Zn-14Al hypereutectic filler
Ultrason. Sonochem., 35 (2017), pp. 61-71
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [43] G.O. Cook, C.D. Sorensen
Overview of transient liquid phase and partial transient liquid phase bonding
J. Mater. Sci., 46 (2011), pp. 5305-5323
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [44] Y. Iino
Partial transient liquid-phase metals layer technique of ceramic-metal bonding
J. Mater. Sci. Lett., 10 (1990), p. 3
[Google Scholar ↗](#)
- [45] M. Xiong, L. Zhang, L. Sun, P. He, W. Long
Effect of CuZnAl particles addition on microstructure of Cu/Sn58Bi/Cu TLP bonding solder joints
Vacuum, 167 (2019), pp. 301-306
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [46] A. Jalali, M. Atapour, M. Shamanian, M. Vahman
Transient liquid phase (TLP) bonding of Ti-6Al-4V/UNS 32750 super duplex stainless steel
J. Manuf. Process., 33 (2018), pp. 194-202
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [47] L. Lan, W. Xuan, J. Wang, C. Li, Z. Ren, J. Yu, J. Peng
Interfacial microstructure of partial transient liquid phase bonding of Si₃N₄ to nickel-base superalloy using Ti/Au/Ni interlayers
Vacuum, 130 (2016), pp. 105-108
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [48] R.B.A. Yarmou Shamsabadi, G. Eisaabadi B
TLP bonding of IN738/MBF20/IN718 system
J. Alloys Compd., 685 (2016), p. 9
[Google Scholar ↗](#)
- [49] R. Sun, Y. Zhu, W. Guo, P. Peng, L. Li, Y. Zhang, J. Fu, F. Li, L. Zhang
Microstructural evolution and thermal stress relaxation of Al₂O₃/1Cr18Ni9Ti brazed joints with nickel foam
Vacuum, 148 (2018), pp. 18-26
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [50] C. Wang, Y.-R. Kim, J.-W. Kim
Numerical analysis of thermal deformation in laser beam heating of a steel plate
J. Mech. Sci. Technol., 31 (2017), pp. 2535-2541
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [51] W.D. Kingery
Factors affecting thermal stress resistance of ceramic materials
J. Am. Ceram. Soc., 38 (1996), p. 13
[Google Scholar ↗](#)

- [52] C.W. Nelson, W.T. Chen
Thermal Stress in Bonded Joints
J. Res. Develop., 23 (1979), p. 10
[Google Scholar ↗](#)
- [53] B.S. Yilbas, A.F.M. Arif, B.J. Abdul Aleem
Laser welding of low carbon steel and thermal stress analysis
Opt Laser. Technol., 42 (2010), pp. 760-768
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [54] P.J. Withers, H.K.D.H. Bhadeshia
Residual stress Part 2 – nature and origins
Mater. Sci. Technol., 17 (2001), p. 10
[Google Scholar ↗](#)
- [55] I.A.A.F.S. Jumbo, A.D. Crocombe, M.M. Abdel Wahab
Thermal residual stress analysis of epoxy bi-material laminates and bonded joints
Int. J. Adhesion Adhes., 30 (2010), p. 15
[Google Scholar ↗](#)
- [56] L.X. Tan Mingming
Influence parameter analysis of residual stress of vacuum brazing joint between glass and metal
Trans. China Weld. Inst., 33 (2012), p. 5
[Google Scholar ↗](#)
- [57] H.S. Bang, H.S. Bang, Y.C. Kim, S.M. Joo
Analysis of residual stress on AH32 butt joint by hybrid CO2 laser-GMA welding
Comput. Mater. Sci., 49 (2010), pp. 217-221
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [58] P.J. Withers

Mapping residual and internal stress in materials by neutron diffraction

Compt. Rendus Phys., 8 (2007), pp. 806-820



[View PDF](#) [View article](#) [Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [59] Y.Y. Santana, J.G. La Barbera-Sosa, M.H. Staia, J. Lesage, E.S. Puchi-Cabrera, D. Chicot, E. Bemporad
Measurement of residual stress in thermal spray coatings by the incremental hole drilling method

Surf. Coating. Technol., 201 (2006), pp. 2092-2098



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [60] N. Murugan, R. Narayanan
Finite element simulation of residual stresses and their measurement by contour method

Mater. Des., 30 (2009), pp. 2067-2071



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [61] S.L. Qu, Y.S. Wang, Y.X. Gai, S. Dong, Y.C. Liang
Residual strains of aluminum alloy characterized by nanoindentation

Trans. Nonferrous Metals Soc. China, 19 (2009), p. 5

[Google Scholar ↗](#)

- [62] J. Lin, N. Ma, Y. Lei, H. Murakawa
Measurement of residual stress in arc welded lap joints by $\cos\alpha$ X-ray diffraction method

J. Mater. Process. Technol., 243 (2017), pp. 387-394



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [63] M.E. Turan, F. Aydin, Y. Sun, M. Cetin
Residual stress measurement by strain gauge and X-ray diffraction method in different shaped rails

Eng. Fail. Anal., 96 (2019), pp. 525-529



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [64] C.S. Choi, H.J. Prask
Residual stress measurements by means of neutron diffraction

Mater. Res. Soc., 166 (1983), p. 6

[Google Scholar](#) ↗

- [65] G.S. Schajer
Measurement of non-uniform residual stresses using the hole-drilling method. Part I—stress calculation procedures

J. Eng. Mater. Technol., 110 (1988), p. 6

[Google Scholar](#) ↗

- [66] B.H. Manning, M.T. Flama
Determination of residual-stress variation with depth by the hole

Drill. Method Exper. Mech. (1985), p. 3

[View in Scopus](#) ↗ [Google Scholar](#) ↗

- [67] A.E.G.S. Suresh
A new method for estimating residual stresses by instrumented sharp indentation

Acta Mater., 46 (1998), p. 13

[Google Scholar](#) ↗

- [68] C.A. Charitidis, D.A. Dragatogiannis, E.P. Koumoulos, I.A. Kartsonakis
Residual stress and deformation mechanism of friction stir welded aluminum alloys by nanoindentation

Mater. Sci. Eng., A, 540 (2012), pp. 226-234



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [69] Y.H. Lee, D. Kwon
Measurement of residual-stress effect by nanoindentation on elastically strained (100) W

Scripta Mater., 49 (2003), pp. 459-465



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

- [70] X. Chen, J. Yan, S. Ren, J. Wei, Q. Wang
Ultrasonic-assisted brazing of SiC ceramic to Ti–6Al–4V alloy using a novel AlSnSiZnMg filler metal

Mater. Lett., 105 (2013), pp. 120-123



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [71] R.S. Chandel, J.X. Zhang, Y.Z. Chen, H.P. Seow
Effect of residual stress on the strength of an alumina–steel joint by partial transient liquid phase (PTLP) brazing

J. Mater. Process. Technol., 122 (2002), p. 6

[Google Scholar ↗](#)

- [72] S. Zhang, M. Yang, M. Jin, W.-C. Huang, T. Lin, P. He, P. Lin, K.-W. Paik
Mechanism of solder joint cracks in anisotropic conductive films bonding and solutions: delaying hot-bar lift-up time and adding silica fillers

Metals (2018), p. 8

[View in Scopus ↗](#) [Google Scholar ↗](#)

- [73] N.S. Rossini, M. Dassisti, K.Y. Benyounis, A.G. Olabi
Methods of measuring residual stresses in components

Mater. Des., 35 (2012), pp. 572-588



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [74] J. Gao, F.L. Duan, C. Yu, W. Meng, L. Liu, G. Ding, C. Zhang, Y. Wang
Electrical insulation of ceramic thin film on metallic aero-engine blade for high temperature sensor applications

Ceram. Int., 42 (2016), pp. 19269-19275



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [75] P.F. Manicone, P. Rossi Iommetti, L. Raffaelli
An overview of zirconia ceramics: basic properties and clinical applications

J. Dent., 35 (2007), pp. 819-826



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [76] L.X. Zhang, L.Z. Wu, D. Liu, J.C. Feng, H.B. Liu
Interface microstructure and mechanical properties of the brazed SiO₂ glass ceramic and 30Cr3 high-tensile steel joint
Mater. Sci. Eng., A, 496 (2008), pp. 393-398
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [77] J.C. Feng, D. Liu, L.X. Zhang, X.C. Lin, P. He
Effects of processing parameters on microstructure and mechanical behavior of SiO₂/Ti–6Al–4V joint brazed with AgCu/Ni interlayer
Mater. Sci. Eng., A, 527 (2010), pp. 1522-1528
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [78] W. Guo, T. Gao, X. Cui, Y. Zhu, P.K. Chu
Interfacial reactions and zigzag groove strengthening of C/C composite and Rene N5 single crystal brazed joint
Ceram. Int., 41 (2015), pp. 11605-11610
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [79] Y. Shen, Z. Li, C. Hao, J. Zhang
A novel approach to brazing C/C composite to Ni-based superalloy using alumina interlayer
J. Eur. Ceram. Soc., 32 (2012), pp. 1769-1774
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [80] Y. Zhang, G. Zou, L. Liu, A. Wu, Z. Sun, Y.N. Zhou
Vacuum brazing of alumina to stainless steel using femtosecond laser patterned periodic surface structure
Mater. Sci. Eng., A, 662 (2016), pp. 178-184
 [View PDF](#) [View article](#) [Crossref ↗](#) [Google Scholar ↗](#)
- [81] J.H. Yang, L.X. Zhang, Z. Sun, J.C. Feng
Brazing SiO₂-BN diphase ceramic with Nb by using multilayer Ti-Ni composite foils
Vacuum, 146 (2017), pp. 179-186

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [82] F.H. Bao, Y. Zhou, J.L. Ren, T.H. North
Interlayer selection and thermal stresses in brazed Si₃N₄- steel joints

Mater. Sci. Technol., 7 (1997), p. 6

[Google Scholar ↗](#)

- [83] J.W. Park, T.W. Eagar
Strain energy release in ceramic-to-metal joints with patterned interlayers

Scripta Mater., 50 (2004), pp. 555-559

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [84] Z. Zhong, Z. Zhou, C. Ge
Brazing of doped graphite to Cu using stress relief interlayers

J. Mater. Process. Technol., 209 (2009), pp. 2662-2670

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [85] T. Zaharinie, R. Moshwan, F. Yusof, M. Hamdi, T. Ariga
Vacuum brazing of sapphire with Inconel 600 using Cu/Ni porous composite interlayer for gas pressure sensor application

Mater. Des., 54 (2014), pp. 375-381

1980-2015

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [86] Y. Qin, J. Feng
Active brazing carbon/carbon composite to TC4 with Cu and Mo composite interlayers

Mater. Sci. Eng., A, 525 (2009), pp. 181-185

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [87] A.A. Shirzadi, Y. Zhu, H.K.D.H. Bhadeshia
Joining ceramics to metals using metallic foam

Mater. Sci. Eng., A, 496 (2008), pp. 501-506



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [88] T. Wang, J. Zhang, C. Liu, G. Wang
Microstructure and mechanical properties of Si₃N₄/42CrMo joints brazed with TiNp modified active filler
Ceram. Int., 40 (2014), pp. 6881-6890



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [89] T. Wang, C. Liu, C. Leinenbach, J. Zhang
Microstructure and strengthening mechanism of Si₃N₄/Invar joint brazed with TiNp-doped filler
Mater. Sci. Eng., A, 650 (2016), pp. 469-477



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [90] Y. Mao, S. Wang, L. Peng, Q. Deng, P. Zhao, B. Guo, Y. Zhang
Brazing of graphite to Cu with Cu₅₀TiH₂+C composite filler
J. Mater. Sci., 51 (2015), pp. 1671-1679
[Google Scholar ↗](#)

- [91] Y.X. Zhao, M.R. Wang, J. Cao, X.G. Song, D.Y. Tang, J.C. Feng
Brazing TC4 alloy to Si₃N₄ ceramic using nano-Si₃N₄ reinforced AgCu composite filler
Mater. Des., 76 (2015), pp. 40-46



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [92] J. Gong, W. Jiang, Q. Fan, H. Chen, S.T. Tu
Finite element modelling of brazed residual stress and its influence factor analysis for stainless steel plate-fin structure
J. Mater. Process. Technol., 209 (2009), pp. 1635-1643



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [93] S. Mishra, S. Pal, S.K. Karak, S. Shah, M. Venakata Nagaraju, A.K. Chakraborty
Effect of thermal cycles and dimensions of the geometry on residual stress of the alumina-kovar joint

IOP Conf. Ser. Mater. Sci. Eng. (2018), p. 338

[Google Scholar ↗](#)

- [94] A. Kar, M. Ghosh, A.K. Ray, A.K. Ray
Effect of interfacial thickness and residual stress on the mechanical property of the alumina–stainless steel
brazed joint interface
Mater. Sci. Eng., A, 498 (2008), pp. 283-288
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [95] H. Dong, Z. Li, X. Song, X. Guo, Y. Luo, T. Bai, S. Wei, H. Zhao, J. Yan, J. Feng
Low-temperature ultrasound-activated joining of ZrO₂ ceramics using Sn-Al-Cu solder
J. Am. Ceram. Soc., 102 (2019), pp. 2272-2277
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [96] T. Nagaoka, Y. Morisada, M. Fukusumi, T. Takemoto
Joint strength of aluminum ultrasonic soldered under liquidus temperature of Sn–Zn hypereutectic solder
J. Mater. Process. Technol., 209 (2009), pp. 5054-5059
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [97] B. Wu, W. Guo, J. He, Z. Xiu, J. Yan
Microstructure evolution of SiC/SiC joints during ultrasonic-assisted air bonding using a Sn–Zn–Al alloy
Ceram. Int., 44 (2018), pp. 1284-1290
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [98] T. Luan, W. Guo, S. Yang, Z. Ma, J. He, J. Yan
Effect of intermetallic compounds on mechanical properties of copper joints ultrasonic-soldered with Sn–Zn
alloy
J. Mater. Process. Technol., 248 (2017), pp. 123-129
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [99] K.S. Kim, K. Suganuma

Sn-Zn low temperature solder

J. Mater. Sci. Mater. Electron., 18 (2007), p. 7

[Crossref ↗](#) [Google Scholar ↗](#)

- [100] L. Zhang, S.B. Xue, L.L. Gao, Z. Sheng, H. Ye, Z.X. Xiao, G. Zeng, Y. Chen, S.L. Yu
Development of Sn–Zn lead-free solders bearing alloying elements

J. Mater. Sci. Mater. Electron., 21 (2009), pp. 1-15

[Crossref ↗](#) [Google Scholar ↗](#)

- [101] D.Q. Yu, H.P. Xie, L. Wang
Investigation of interfacial microstructure and wetting property of newly developed Sn–Zn–Cu solders with Cu substrate

J. Alloys Compd., 385 (2004), pp. 119-125



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [102] H. Wang, S. Xue, F. Zhao, W. Chen
Effects of Ga, Al, Ag, and Ce multi-additions on the properties of Sn–9Zn lead-free solder

J. Mater. Sci. Mater. Electron., 21 (2009), pp. 111-119

[Google Scholar ↗](#)

- [103] Jeong K. Hong, Pingsha Dong
Residual Stress Relief in Post-Weld Heat Treatment Proceedings of the ASME Pressure Vessel and Piping Conference

(2008), p. 5

[Crossref ↗](#) [Google Scholar ↗](#)

- [104] Z. Yan, H. Zhang, J. Duan, F. Liu, G. Wang
Effect of post-weld heat treatment on mechanical characteristics of AZ31 magnesium alloy welded joints

J. Wuhan Univ. Technol.-Materials Sci. Ed., 32 (2017), pp. 1205-1212

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [105] Z. Hu, S. Yuan, X. Wang, G. Liu, Y. Huang
Effect of post-weld heat treatment on the microstructure and plastic deformation behavior of friction stir welded 2024
Mater. Des., 32 (2011), pp. 5055-5060
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [106] C.N. Niu, X.G. Song, S.P. Hu, G.Z. Lu, Z.B. Chen, G.D. Wang
Effects of brazing temperature and post weld heat treatment on 7075 alloy brazed joints
J. Mater. Process. Technol., 266 (2019), pp. 363-372
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
-

Cited by (103)

Eutectic high-entropy alloys and their applications in materials processing engineering: A review

2024, Journal of Materials Science and Technology

[Show abstract](#) ✓

Characteristics, applications and perspective of high entropy alloys for interfacial joining: A review

2024, Journal of Manufacturing Processes

[Show abstract](#) ✓

High shear strength Kovar/AlN joints brazed with AgCuTi/Cu/AgCuTi sandwich composite filler

2023, Materials Science and Engineering A

[Show abstract](#) ✓

The regulation strategy for releasing residual stress in ceramic-metal brazed joints

2023, Journal of Manufacturing Processes

[Show abstract](#) ✓

Brazing C/C composites to DD3 alloy with a novel Ag–Cr active braze

2022, Ceramics International

[Show abstract](#) ✓

Recent development of the novel riveting processes ↗

2021, International Journal of Advanced Manufacturing Technology



[View all citing articles on Scopus](#) ↗

[View Abstract](#)

© 2021 Elsevier Ltd and Techna Group S.r.l. All rights reserved.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.